Floating Satellite System (FloatSat)

Exercise 7: Operation Modes

• Exercise 7: Operation Modes

>> The "OnlineFloatSat" source project contains four main threads which communicate with each other using topics:

• The first thread is the "Sensors" thread runs at 5ms time interval. This thread collects sensors information, process them and then publishes the data stored in sensorData `struct` into SensorDataTopic to be received by other threads.
*** Please refer to „topics.h" for sSensorData `struct`.**

• The second thread is the "Telecommand" thread runs upon receiving data. This thread decodes the received telecommand messages, and publishes the data stored in TelecommandData `struct` into TelecommandDataTopic to be received by other threads only if the message is valid.
• The message body of the telecommand should have the following format "$Xdata#" where:
"$" is the Message Start Character [1 byte].
X is the Message ID Character [1 byte].
data are ASCII characters which correspond to a valid floating point number [0 –12 bytes].
"#" is the Message End Character [1 byte].
*** Please refer to „topics.h" for "sTelecommandData" `struct`.**
*** Please refer to „Telecommand.h" for Message ID Characters.**

• The third thread is the "Telemetry" thread runs at 1000ms time interval. This thread publishes the sensors information to the ground station.

• The fourth thread is the "Modes" thread runs at 10ms time interval. This thread runs different operation mode for the FloatSat. There are four modes available: *`Standby_Mode`*, *`Velocity_Mode`*, *`Position_Mode`*, *`MotorSpeed_Mode`*.
*** Please refer to „Telecommand.h" for corresponding Message ID Character.**
*** Please refer to „Modes.h" for Operation_Mode `enum`.**

• Exercise 7: Operation Modes

>> Establishing connection with the Ground Station :

• In order operate the FloatSat, we need to establish connection with it from the ground station (PC) through available Bluetooth dongle. Please add the device for your corresponding Bluetooth module in your FloatSat with default Pin code of "1234". Please check which COM port, Windows has assign to the added device and use it in your "HTerm" window with default baud rate of 115200.

• You can enable/disable telemetry data by sending the corresponding telecommand „$T1#"/"$T0#".

>> Sensors Calibration :

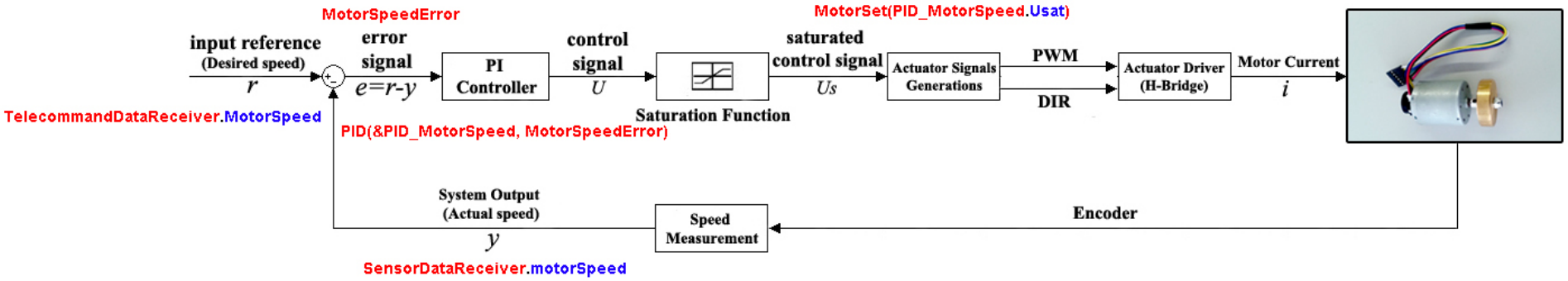
• Gyroscope Initialization and Calibration process is performed at the system startup, please make sure that the system is at standstill when you switch it on.

• The Magnetometer needs to be calibrated for the magnetic distortion around your FloatSat. Please initiate magnetometer calibration process by switching the AHRS_mode to `Mag_Cal` mode upon sending the corresponding telecommand.
*** Please refer to „Telecommand.h" for corresponding Message ID Character.**
*** Please refer to „Sensors.h" for corresponding AHRS_Mode `enum`.**

• Exercise 7: Operation Modes

>> Operation Modes:

• A PI controller for controlling the Motor Speed needs to be implemented inside `void MotorSpeedMode()` as following:

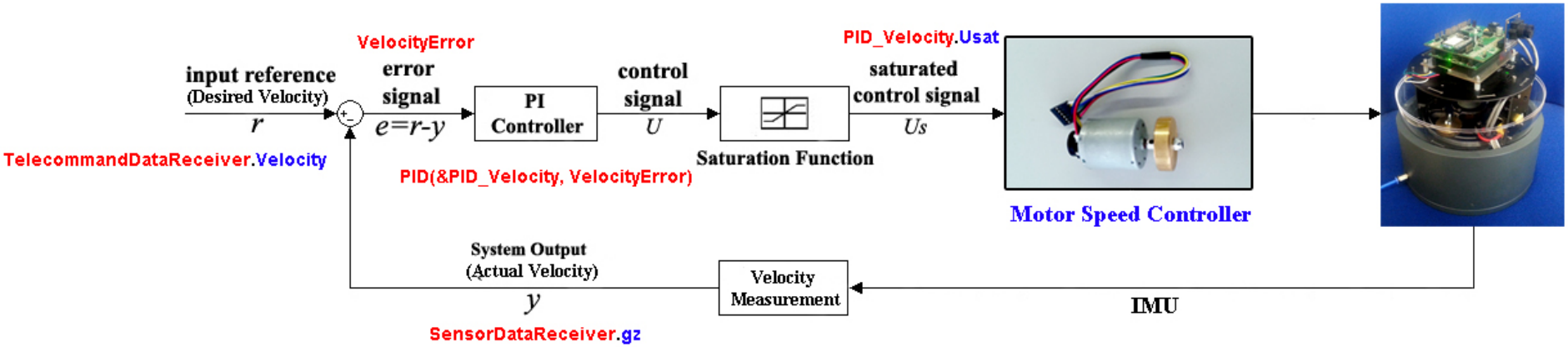


- **Tune the PI controller online to have a good performance characteristics of a very fast response with little overshoot by sending corresponding telecommands.**

• Exercise 7: Operation Modes

>> Operation Modes:

- **A PI controller for controlling the FloatSat Angular Velocity needs to be implemented inside `void VelocityMode()` as following:**

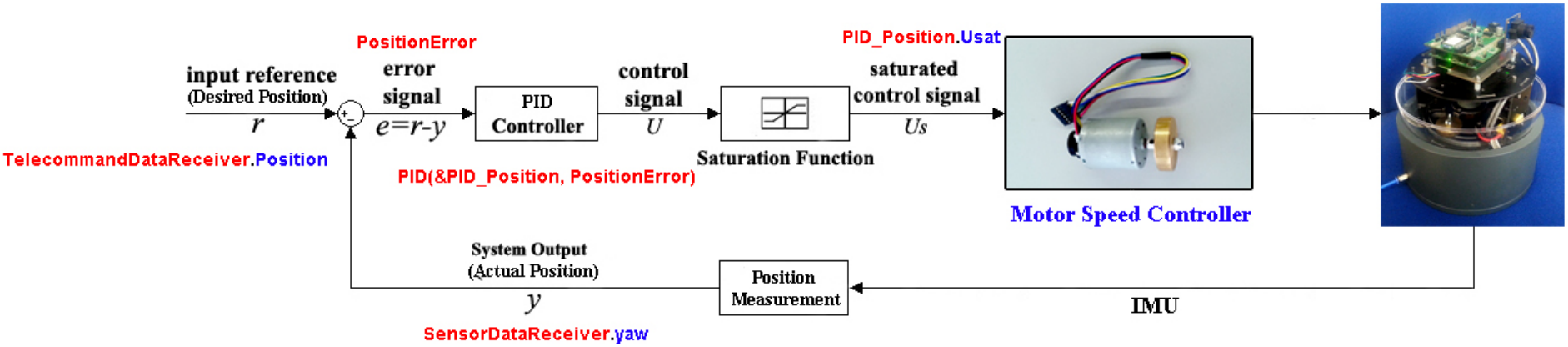


- **Tune the PI controller online to have a good performance characteristics of fast response with very little overshoot by sending corresponding telecommands.**

• Exercise 7: Operation Modes

>> Operation Modes:

- **A PID controller for controlling the FloatSat Angular Position needs to be implemented inside `void PositionMode()` as following:**



- **Tune the PID controller online to have a good performance characteristics of response with no overshoot by sending corresponding telecommands.**